

Replication codes for Fernández-Villaverde, Hurtado and Nuño (2025): "Corrigendum: Financial Frictions and the Wealth Distribution".

Overview

The code in this replication package runs the model from Fernández-Villaverde, Hurtado and Nuño (2025): "Corrigendum: Financial Frictions and the Wealth Distribution" using Matlab. It can be used to solve the model and generate all the graphs in the corrigendum. The replicator should expect the code to run for about 10 hours.

Data Availability and Provenance Statements

The only data used in the paper is the series of GDP for the USA, as downloaded from the FRED website of the Federal Reserve. The "Data" folder contains said data as used in the paper, and the code with which a standard bandpass filter is applied before feeding the results to the model. The main XLS/CSV file contains hyperlinks to the download page. The file `a2_launch.m` runs all the preparation codes in this folder.

Statement about Rights

- I certify that the authors of the manuscript have legitimate access to and permission to use the data used in this manuscript.

Summary of Availability

- All data **are** publicly available.

Software Requirements

- Matlab (code checked on various versions from r2016 to r2021)

Controlled Randomness

For reproducibility purposes, seeds of the pseudorandom number generators have been set in the codes.

- Random seed is set at line 58 of program `b1_parameters.m`

Memory, Runtime, Storage Requirements

Approximate time needed to reproduce the analyses on a standard 2025 desktop machine: 7-10 hours

Approximate storage space needed: 4 GB (commenting the 3 "save `z_finalworkspace`" lines at the end of `a2_launch.m` reduces this to 500 MB)

The code was last run on a 4-core Intel I7- 7770K computer with 32GB RAM (which took less than 10 hours to run) and on a 32-core AMD Threadripper W3970X with 128GB RAM (which took slightly over 7 hours to run).

Description of programs/code

<code>a2_launch.m</code>	Launches everything, including the preliminary stages (solving the steady state, and solving a version of the model with a linear regression, to be used as starting value in the fully non-linear version of the model) and the generation of the figures
<code>b1_parameters.m</code>	Sets all the parameters of the model
<code>b2_K1m.m</code>	Runs the main loop
<code>b3_HJB.m</code>	Solves the HJB equation
<code>b5_KFE.m</code>	Solves the KF equation
<code>b5_use_parallel.m</code>	Use the parallel toolbox if there is a licence available and at least 12GB of RAM
<code>b7_PLM.m</code>	Updates the PLM using a neural network
<code>b7_PLM_iter.m</code>	Subroutine of <code>b7_PLM.m</code> in charge of running the line search optimization
<code>b9_plot.m</code>	Creates basic diagnostic plots to assess convergence and performance
<code>f1_NN_loss.m</code>	Computes the loss function of the neural network
<code>f2_NN_eval.m</code>	Evaluates the output of the neural network for a given set of data
<code>f5_NN_gradient.m</code>	Computes the gradient of the neural network
<code>f8_KFE_sim.m</code>	Simulates the dynamics of the distribution using the KF

After solving the model, `a2_launch.m` will go into the figures subfolder and execute another set of .m files that generate the different figures in the paper.

Instructions to Replicators

Running `a2_launch.m` should be enough to get everything done. It will first go into the folders for the preliminary stages (steadystate and model_LR; these parts should take between 90 and 120 minutes), then use that to solve the full version of the model (most of the computing time is spent here), then go into the figures subfolder to plot all the figures (this takes only a few minutes).

List of tables and programs

The provided code reproduces all tables and figures in the paper. The relevant files are in the figures subfolder.

Figure	Program	Output file	Notes
Figure 1	<code>c2_PLM_NPLM_phase.m</code>	<code>g91_phase</code>	Change <code>nval_B</code> parameter and solve the model again to get the different panels
Figure 2	<code>c4_errors_zone.m</code>	<code>g92_zone_3D</code>	
Figure 3	<code>c3_sss.m</code>	<code>g93_SSS</code>	
Figure 4	<code>c5_IRF.m</code>	<code>g94_IRF</code>	
Figure 5	<code>c4_errors_zone.m</code>	<code>g95_PLMerrors</code>	

References

Fernández-Villaverde, Jesús, Samuel Hurtado, and Galo Nuño. "Financial Frictions and the Wealth Distribution." *Econometrica*, 91, .no 3, (Econometric Society: 2023), 869-901. <https://doi.org/10.3982/ECTA18180>